

FOSSDAQ

ai-8x-0to3v3

Build Manual

8-channel analog input card (0–3.3 V) on TS-35 DIN rail

Project: FOSSDAQ

Module: ai-8x-0to3v3

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Hardware version: v1.0 (July 2026)

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1 Basic data

*The ai-8x-0to3v3 card is the first fully developed measurement module of the FOSSDAQ system. It accepts up to **eight** actively supplied sensors with a 0–3.3 V signal output — typical pressure or flow sensors — supplies them with stabilized 5 V and delivers the readings to the Raspberry Pi via USB.*

Property	Value
Function	8-channel analog input card for 5 V sensors with 0–3.3 V output
Controller	Arduino Nano 33 BLE Rev2 (alternative: Arduino Nano R4)
ADC	12-bit, analog pins A0–A7
Supply	24 V DC bus input; local 5 V via LM2596S (input range 10–35 V DC)
Sensor supply	5 V from the local rail, up to 2 A available (a few 100 mA used)
Signal range	0–3.3 V with the Nano 33 BLE Rev2; 0–5 V when built with the Nano R4
Data	USB to the Raspberry Pi (data only)
Mounting	TS-35 DIN rail, springy printed latch pin
Protection class	SELV; common ground with the bus, no galvanic isolation

Only sensors with 5 V supply and 0–3.3 V output

The sensors are powered with 5 V, but their signal goes directly to the analog inputs of the Nano 33 BLE Rev2. These are 3.3 V pins and **not** 5 V tolerant. Only use sensors whose signal output **never exceeds** 0–3.3 V (for example 0.5–2.5 V types). **Not suitable** are the common ratiometric 0.5–4.5 V sensors: their output reaches 4.5 V and would overdrive the input — they would need a voltage divider.

Variant: if the card is built with the **Arduino Nano R4** instead (5 V logic, 14-bit ADC), the analog inputs accept **0–5 V** — ratiometric 0.5–4.5 V sensors then connect directly, without a voltage divider.

2 Components

Parts needed for one card (see ../bom/ for the machine-readable list):

Qty	Part	Order code	Supplier	Remark
1	Arduino Nano 33 BLE Rev2	ARD NANO 33BLE2	Reichelt	nRF52840, 3.3 V logic; alternative: Nano R4 (USB-C, 5 V logic — accepts 0–5 V inputs)
2	Breadboard/perfboard 80 × 20 mm, double-sided	LR-DS-28	Reichelt	1 × sensor terminal carrier, 1 × strip for the bus terminal
4	Spring loaded terminal AST 025-06 (6-pin)	AST 025-06	Reichelt	sensor connections, 2 sensors per terminal
1	Spring loaded terminal AST 025-04 (4-pin)	AST 025-04	Reichelt	24 V bus input + pass-through
1	DC/DC step-down converter LM2596S with display	DEBO DCDC DOWN 5	Reichelt	Joy-IT module with potentiometer, trim to 5.0 V before use
1	USB-A ↔ Micro-USB-B data cable, 1 m	—	free choice	data link to the Pi (USB-C cable for the Nano R4)
≈ 4 m	Stranded wire H05V-K 1 × 0.75 mm ² , red / black / green	—	e.g. Conrad	wire colors freely selectable; recommended: red +24 V/+5 V, black 0 V, green sensor signals A0–A7
≥ 15	Screws M2.3 × 6 mm, self-tapping, Torx	ASIN B0G6YNG9CW	Amazon	cover screws, thread directly into PLA; sets usually contain more anyway
—	PLA filament	—	free choice	PETG as alternative

Tools: soldering iron with fine tip, lead-free solder, fine saw for the perfboard, side cutter, wire stripper, multimeter, T6 Torx bit, scalpel.

3 Circuit

3.1 Schematic

The circuit is deliberately minimal: bus input terminal, step-down converter, Arduino and sensor terminals. KiCad sources: `../hardware/kicad/`.

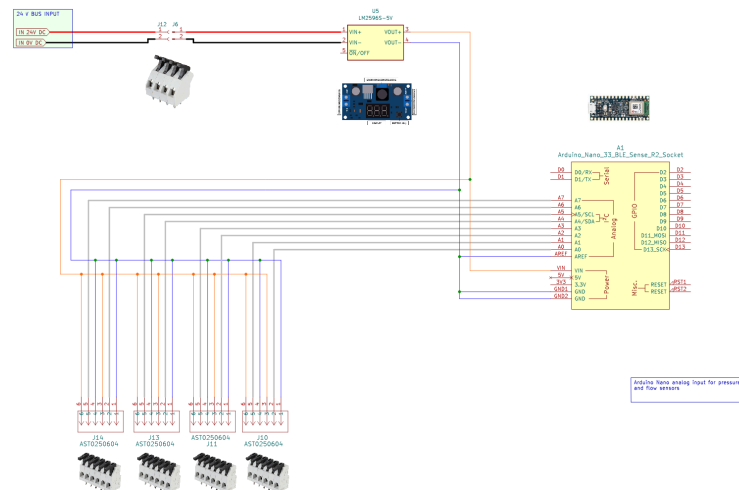


Figure 1: Schematic of the ai-8x-0to3v3 card.

Section	Part	Function
Input terminal	Spring loaded terminal AST 025-04	24 V and 0 V from the bus; the second terminal pair passes the bus on
DC/DC converter	LM2596S module	converts 24 V to a stable 5 V for Arduino and sensors
Arduino	Nano 33 BLE Rev2	digitizes the sensor signals and sends them to the Pi via USB
Sensor terminals	four 6-pin spring loaded terminals AST 025-06	two sensors per terminal, up to eight in total

3.2 Terminal pinout (two sensors per terminal)

Each 6-pole spring terminal serves **two** sensors with three wires each. The poles are arranged vertically, **pole 1 on top, pole 6 at the bottom**. +5 V and 0 V run through all terminals as common supply rails; only the signal wire goes individually to the Arduino:

Pole	Signal	Suggested color
1 (top)	+5 V for sensor A	red
2	signal A, 0–3.3 V → Nano	green
3	0 V / GND for sensor A	black
4	+5 V for sensor B	red
5	signal B, 0–3.3 V → Nano	green
6 (bottom)	0 V / GND for sensor B	black

3.3 Current budget: are eight sensors fine?

Arduino and sensors share the 5 V output of the LM2596S, so this module is the limiting element:

Item	Value
LM2596S, 5 V output	up to 2 A continuous
Arduino Nano 33 BLE	~ 60 mA
Eight sensors at 10–30 mA each	≈ 80–240 mA
Total	≈ 140–300 mA

The total load of roughly 140–300 mA is far below the 2 A of the module. **Eight sensors are no problem** — plenty of reserve remains. Only power-hungry sensors drawing several 100 mA each would approach the limit.

4 Enclosure

The enclosure follows the same principle as the pi-base module: a main body with integrated DIN-rail recess and latch pin, a screwed side cover and a slim form factor for row mounting.

Part	FreeCAD file	Function
Main body	AI-Gehaeuse+Slate.FCStd	holds the carrier board with terminals and the Arduino; DIN-rail recess on the back
Cover	AI-Deckel_v2.FCStd	side cover, screwed on
Latch pin	AI-Stift.FCStd	springy pin that locks the enclosure onto the DIN rail

Ready-to-print files (3MF and STL) are in ../hardware/print/; the editable FreeCAD sources are in ../hardware/cad/.

4.1 Main body

The main body holds the carrier board with the terminals and the Arduino in a slide-in slot. Ventilation grids sit on top and bottom.

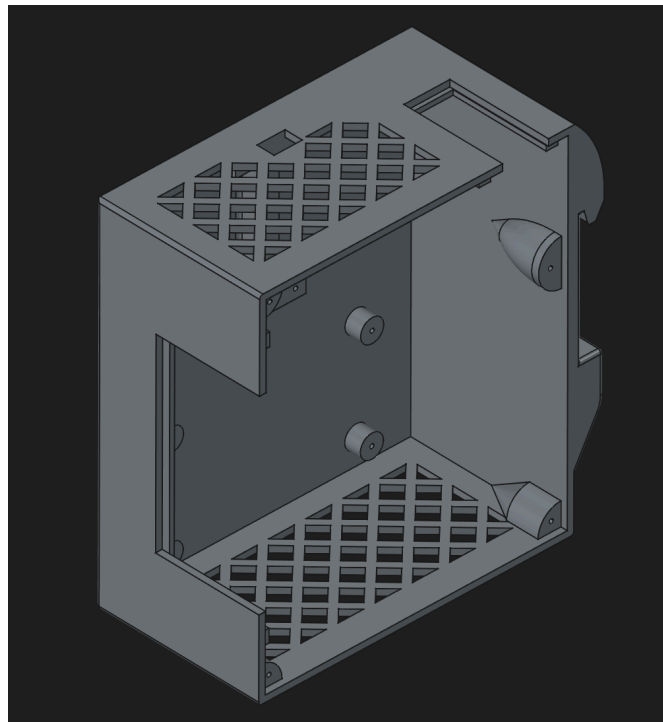


Figure 2: ai-8x-0to3v3 main body, open.

4.2 Cover

The cover closes the enclosure on the side and is fixed with four screws.

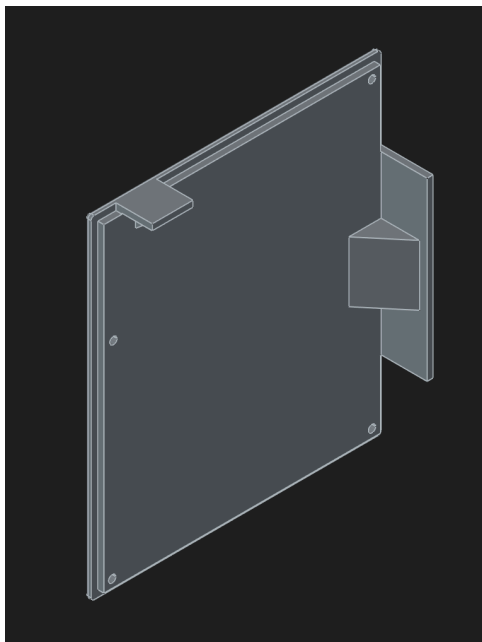


Figure 3: Side cover of the ai-8x-0to3v3 enclosure.

4.3 DIN-rail latch pin

The pin is identical in design to the pi-base pin: a separately printed springy element. Pushing the enclosure onto the rail makes it snap in; to release it, pull the pin back with a flat screwdriver.

5 3D printing

The print process is identical to the pi-base module.

Parameter	Value
Filament	PLA (matte), alternatively PETG
Nozzle / bed (PLA)	210 °C / 60 °C
Supports	not required
Attention	thin walls of the slide-in slot — check the first layer carefully

6 24 V bus terminal assembly

The 24 V feed sits on its own narrow board strip, cut from an 80 × 20 mm perfboard (2.54 mm pitch). The bus terminal can thus be populated independently and then slid into the guide of the enclosure.

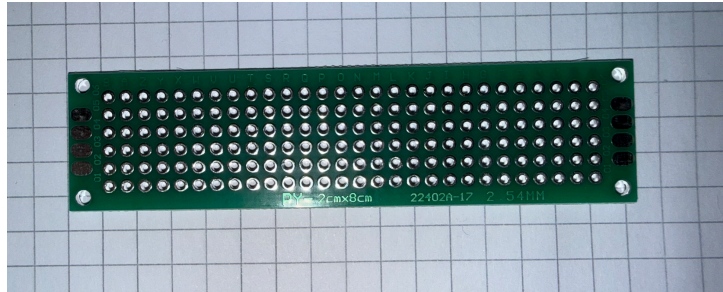


Figure 4: Double-sided 80 × 20 mm perfboard, raw.

- 1. Cut the strip.** Saw the marked narrow strip off the perfboard along a hole row and deburr the cut edge.
- 2. Mount and solder the 4-pole terminal.** Place the AST 025-04 on the strip and solder it. Bridge +24 V and 0 V with one red and one black wire jumper to the second terminal pair, so the next card can be fed from the same bus.
- 3. Slide it in.** Slide the finished terminal carrier into the side guide of the enclosure.

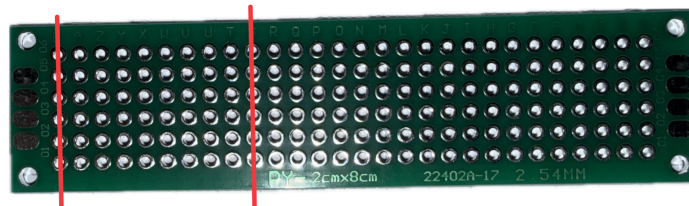


Figure 5: Cut line (red): strip for the 24 V bus terminal.

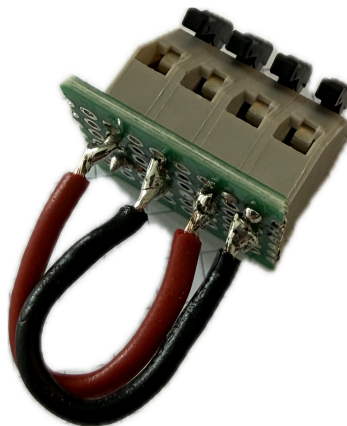


Figure 6: Soldered 4-pole bus terminal AST 025-04 with the pass-through bridges: +24 V in red, 0 V in black.

7 Sensor terminal carrier board

Each card uses a single double-sided 80 × 20 mm perfboard (2.54 mm pitch), cut to the enclosure's inner dimension. It carries only the **four 6-pole spring terminals** for the sensors. All wiring and bridging is done from the **rear** side.

Leave a 2 mm gap between terminals and board

The terminals are **not** soldered flush onto the board but with a gap of about **2 mm**. The enclosure wall grips into this gap when the board is slid in: the carrier board sits behind the wall while the terminals protrude in front of it. Without the gap the board cannot be inserted.

- 1. Set the terminals with a 2 mm gap.** Place the four 6-pole terminals so that a gap of about 2 mm remains between terminal body and board, then solder from the rear. Poles are vertical, **pole 1 on top, pole 6 at the bottom**; each terminal serves two sensors according to the pinout table above.
- 2. Bridge the supply rails.** On the rear side, connect the +5 V rail (red) and the 0 V rail (black) across **all** terminals with short wire bridges, so every sensor sees the same supply. Only the signal poles stay individual.
- 3. Prepare the signal wires.** Cut the eight green signal wires to length for the later connection to A0–A7 of the Nano.

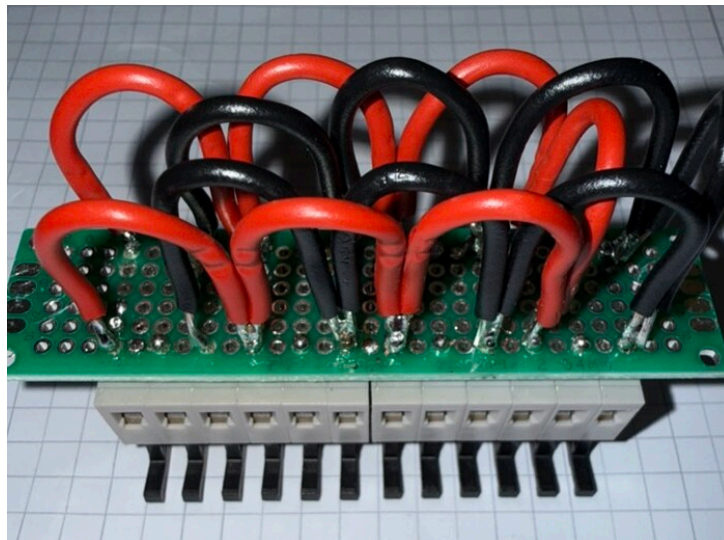


Figure 7: Rear of the sensor terminals with the bridged supply rails: +5 V in red, 0 V in black.

8 Soldering the Arduino Nano

The Arduino Nano is soldered **directly** via its pin headers. The relevant pins:

Arduino pin	Signal	Connected to
VIN	+5 V from DC/DC	5 V rail from the LM2596S output
GND	ground	common 0 V
AREF	analog reference	soldered to the common 0 V as reference potential
A0–A7	analog inputs	signals of the eight sensors, 0–3.3 V
5 V output pin	not used	sensors are fed directly from the 5 V rail

9 Mechanical assembly

With the 24 V terminal and the carrier board populated and the Nano wired, everything comes together in the enclosure:

- 1. Deburr the printed parts.** Remove sharp edges in the slide-in slot and at the terminal cut-outs with a scalpel.
- 2. Install the DC/DC converter and trim it to 5.0 V.** Mount the LM2596S module on a socket strip so the trim potentiometer stays accessible. Connect its input to +24 V and 0 V of the bus terminal. **Before** connecting the Nano: apply 24 V, trim the output to exactly **5.0 V** with a multimeter, then switch the 24 V off. Continuity check: no short 24 V → 0 V and 5 V → 0 V.
- 3. Connect the Nano.** Solder the 5 V rail to VIN and GND to the common ground.
- 4. Slide in the carrier board.** Push the populated carrier board into the slot until it sits flush; the enclosure wall grips into the 2 mm gap between board and terminals.
- 5. Route the wires.** Lead the eight green signal wires from the terminals to A0–A7 and route the +5 V and 0 V rails to the Nano. Keep the runs short and tidy.
- 6. Screw on the cover and insert the pin.** Fix the cover with four M2.3 screws — do not overtighten, PLA cracks easily. Insert the latch pin with its spring tab pointing up.
- 7. Check terminals and USB.** Test the spring travel of all terminals, plug in the USB cable to the Pi and check its fit.

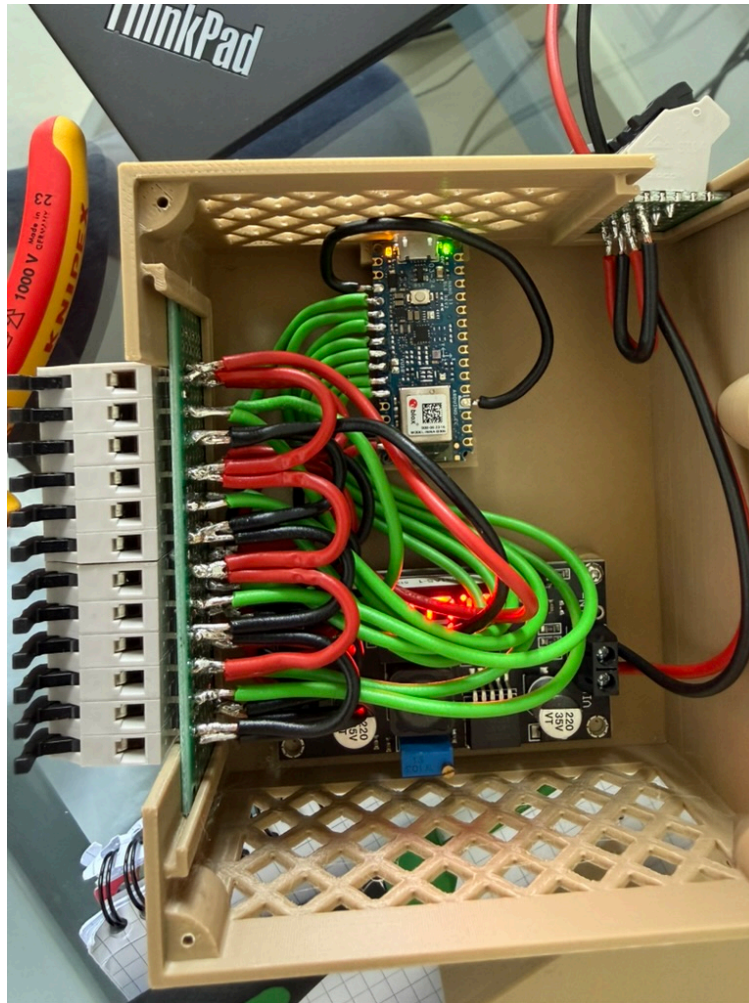


Figure 8: Fully wired card in the open enclosure: carrier board with sensor terminals, signal wires to the Arduino, LM2596S module.

10 Checks before commissioning

Firmware and Pi-side software are defined by the project supervision. Before handover, check the hardware:

- 1. Visual inspection of all solder joints** — clean, shiny joints, no cold joints or bridges; check the orientation of polarized parts and the fit of the Arduino.
- 2. Continuity check.** No short between +24 V and 0 V, and none on the 5 V rail after the DC/DC converter.
- 3. Current-limited 24 V bench supply.** Connect the card without the Arduino and watch the current: it must settle at a stable low value. If the current limit trips immediately, there is a short.
- 4. Measure and trim the 5 V output to 5.00 V**, then secure the potentiometer with a drop of threadlocker.
- 5. Mechanical checks.** Enclosure snaps onto a 35 mm DIN rail and releases again; spring terminals hold a 1.0 mm² test wire and release it cleanly.

11 Finished module



Figure 9: Bottom view with ventilation grid and terminals.

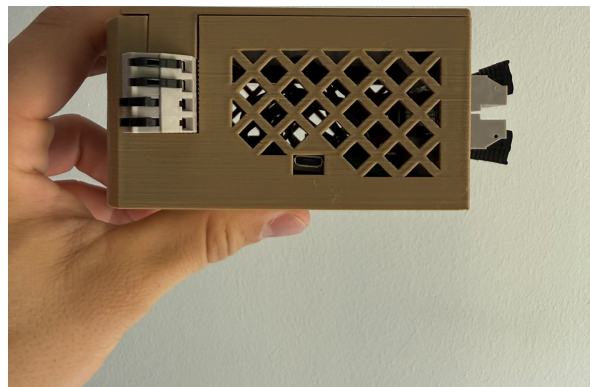


Figure 10: Top view with the 24 V bus terminal, the sensor terminals and the Micro-USB-B socket.